7.4.1 Mathematical statement of the first law

The mathematical statement of the first law of thermodynamics is

$$\Delta U = q + w \quad \text{--------- (7.7)}$$

Case 1 : For a cyclic process involving isothermal expansion of an ideal gas,

$$\Delta U = 0.$$

eqn (7.7) $\Rightarrow \therefore q = -w$

In other words, during a cyclic process, the amount of heat absorbed by the system is equal to work done by the system.

Case 2 : For an isochoric process (no change in volume) there is no work of expansion. i.e. $\Delta V = 0$

$$\Delta U = q + w$$

$$= q - P\Delta V$$

$$\Delta V = 0$$

$$\Delta U = q_v$$

In other words, during an isochoric process, the amount of heat supplied to the system is converted to its internal energy.

Case 3 : For an adiabatic process there is no change in heat. i.e. q= 0. Hence

$$q = 0$$

eqn (7.7) $\Rightarrow \Delta U = w$

In other words, in an adiabatic process, the decrease in internal energy is exactly equal to the work done by the system on its surroundings.

Case 4 : For an isobaric process. There is no change in the pressure. P remains constant. Hence

$$\Delta U = q + w$$

$$\Delta U = q - P\,\Delta V$$

In other words, in an isobaric process a part of heat absorbed by the system is used for P-V expansion work and the remaining is added to the internal energy of the system.

Problem: 7.1

A gas contained in a cylinder fitted with a frictionless piston expands against a constant external pressure of 1 atm from a volume of 5 litres to a volume of 10 litres. In doing so it absorbs 400 J of thermal energy from its surroundings. Determine the change in internal energy of system.

Solution:

Given data q = 400 J V_1=5L V_2 = 10L

Δu = q-w (heat is given to the system (+q); work is done by the system(-w)

$$\Delta u = q - PdV$$

$$= 400\ J - 1\ atm\ (10-5)L$$

$$= 400\ J - 5\ atm\ L$$

[$\therefore$1L atm = 101.33 J]

$$= 400\ J - 5 \times 101.33\ J$$

$$= 400\ J - 506.65\ J$$

$$= -106.65\ J$$

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